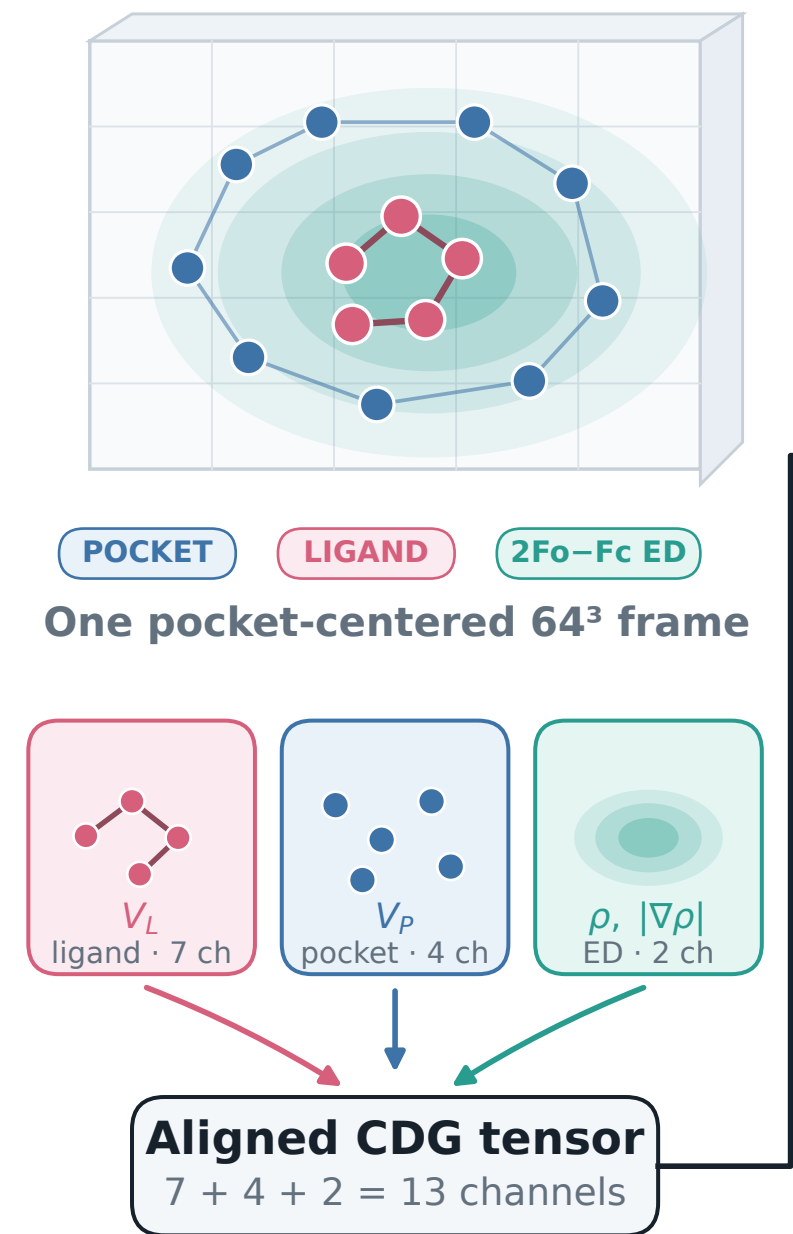


Single-encoder CDG pipeline

Pocket coordinates, ligand coordinates, and experimental electron density are learned jointly in one spatial encoder

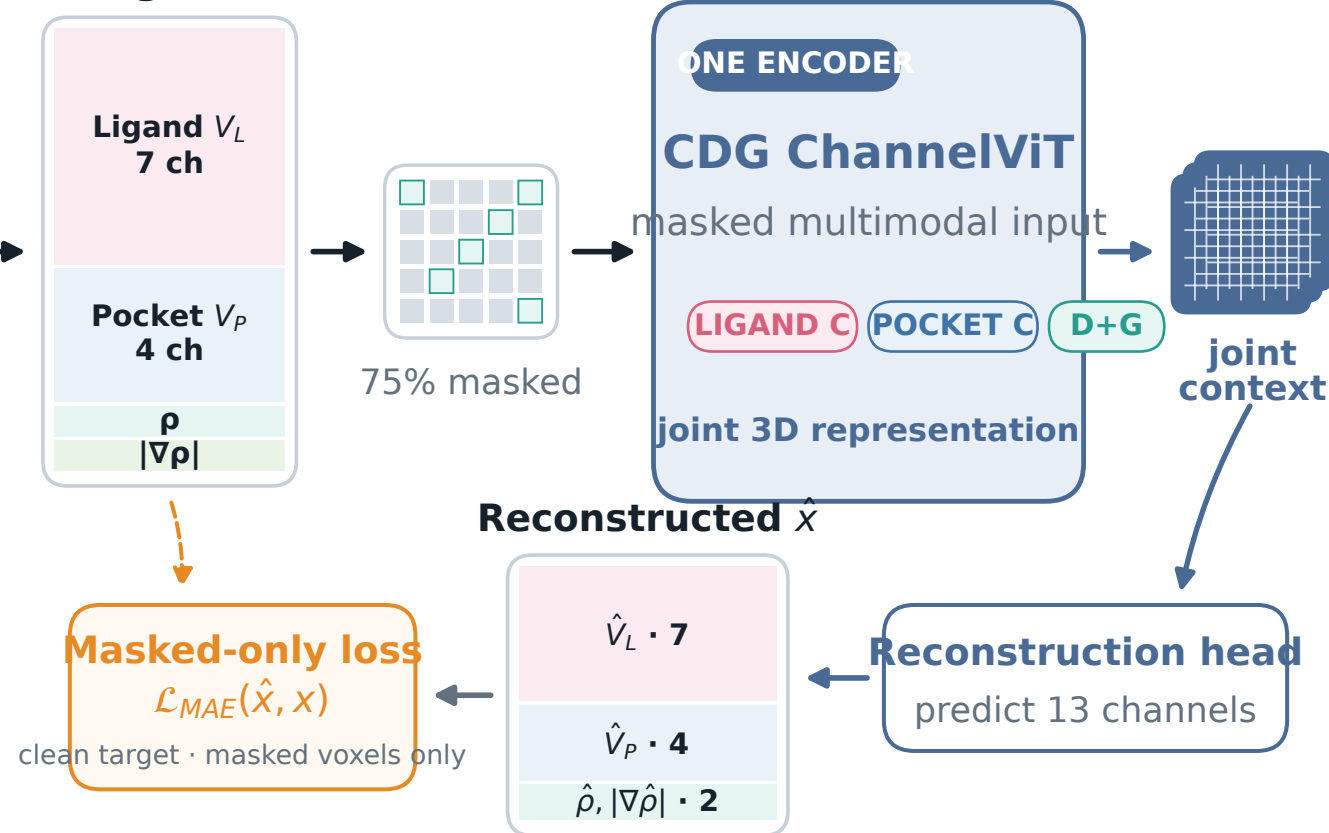
(a) Aligned input data



C: coordinates · D: density · G: density gradient

(b) Joint CDG pre-training

Clean target $x \cdot 13$ ch



One encoder reconstructs ligand, pocket, and ED together

The learned context preserves their shared 3D geometry

(c) High-level VoxBind fusion

FUSION TEST

Original VoxBind stream

